

# EXAMPLES

$$\begin{array}{c} \text{blue arrow} \rightarrow (1 \quad 4) \end{array} \quad \begin{array}{c} \text{red arrow} \rightarrow \\ \text{red arrow} \rightarrow \end{array} \begin{pmatrix} -3 & 2 \\ 1 & 1 \\ A & F \end{pmatrix}$$

Diagram illustrating a transformation from a single row vector  $(1 \quad 4)$  to a matrix  $\begin{pmatrix} -3 & 2 \\ 1 & 1 \\ A & F \end{pmatrix}$ . Blue arrows point to the source vector and the target matrix. Red arrows indicate the mapping of the source elements to the target matrix rows.



$$\begin{array}{c} \text{blue arrow} \rightarrow (5 \quad 1 \quad 21) \end{array} \quad \begin{array}{c} \text{red arrow} \rightarrow \\ \text{red arrow} \rightarrow \\ \text{red arrow} \rightarrow \end{array} \begin{pmatrix} 0 & -1 & 2 \\ 1 & -2 & 4 \\ 1 & 1 & 0 \\ V & I & N \end{pmatrix}$$

Diagram illustrating a transformation from a single row vector  $(5 \quad 1 \quad 21)$  to a matrix  $\begin{pmatrix} 0 & -1 & 2 \\ 1 & -2 & 4 \\ 1 & 1 & 0 \\ V & I & N \end{pmatrix}$ . Blue arrows point to the source vector and the target matrix. Red arrows indicate the mapping of the source elements to the target matrix rows.

# ENIGME

$$\begin{pmatrix} 3 & 12 & 15 & 19 & 5 \end{pmatrix} \begin{pmatrix} 10 & -1 & -5 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 \\ -3 & 0 & 4 & -1 \end{pmatrix}$$

Diagram illustrating a transformation from a single row vector  $(3 \quad 12 \quad 15 \quad 19 \quad 5)$  to a matrix  $\begin{pmatrix} 10 & -1 & -5 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 \\ -3 & 0 & 4 & -1 \end{pmatrix}$ .